MEDICAL INSURANCE COST PREDICTOR: A REGRESSION BASED APPROACH

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Abstract—Medical insurance costs can vary considerably across individuals because they are associated with several demographic, physical, and lifestyle characteristics. Estimating these costs from historical records is a useful application of supervised machine learning and regression analysis. This paper presents the design of a Medical Insurance Cost Predictor that uses features such as age, sex, body mass index (BMI), number of children, smoking status, and residential region to estimate medical insurance charges. The proposed workflow covers dataset inspection, data cleaning, treatment of numerical and categorical variables, exploratory data analysis, model development, and evaluation. Two regression techniques are considered: Linear Regression as a baseline model and Random Forest Regression to represent possible non-linear relationships and interactions among the input features. Their performance is intended to be assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2), using a separate test set. The project also proposes a Streamlit-based dashboard through which a user can enter the required details and view a predicted charge, while data visualizations can help communicate patterns in the dataset. The study emphasizes reproducible preprocessing, consistent feature handling during training and prediction, and transparent reporting of model limitations. The commonly used Medical Cost Personal Dataset contains United States regional categories and charges in its source currency; therefore, it cannot by itself support validated predictions for Indian regions or Indian-rupee costs. An India-focused implementation would require a suitable Indian dataset and retraining. Since final model evaluation outputs have not yet been supplied, this paper describes the proposed methodology and does not claim measured predictive performance. The intended outcome is an academic prototype that demonstrates how regression models can be applied to insurance-cost estimation, rather than an official insurance quote or a substitute for professional financial or insurance advice.

Index Terms—Medical insurance, cost prediction, regression, machine learning, exploratory data analysis, Random Forest, Streamlit.

I. INTRODUCTION

Medical insurance plays an important role in helping individuals and families manage the financial impact of healthcare needs. Medical expenses can vary considerably depending on a person's health condition, age, lifestyle, treatment requirements, and insurance policy. The amount charged to a policyholder may also depend on several personal and demographic factors, including age, body mass index (BMI), smoking habits, number of dependents, and residential location. Since these characteristics may influence medical expenditure in different ways, estimating insurance costs by considering only one factor may provide an incomplete understanding. Machine-learning and regression-based approaches have therefore been explored to identify patterns in healthcare expenditure and estimate medical insurance costs [7], [8], [10]–[13]. Research has also examined explainable prediction methods to understand how individual input variables contribute to estimated insurance costs [1].

Healthcare expenditure has become an important financial consideration because medical treatment may involve consultation fees, diagnostic tests, medicines, hospitalization, and follow-up care. These expenses can differ from one individual to another based on personal circumstances and healthcare requirements. Understanding the factors associated with medical insurance charges can help individuals become more aware of possible financial requirements. An analysis of historical insurance records can provide useful information about relationships between personal characteristics and recorded medical charges. Machine learning provides a systematic way to analyze such records and identify patterns that may be difficult to recognize through simple manual analysis.

Earlier studies have investigated different approaches to medical insurance cost prediction. These include conventional regression models, ensemble learning methods, and computational-intelligence techniques for estimating healthcare expenditure and insurance premiums [2], [3], [7]–[12]. Linear Regression is commonly used as a baseline algorithm because it is simple to implement and provides an interpretable relationship between input variables and a continuous target value. However, a linear model may not represent all complex relationships among the features. Random Forest Regression is an ensemble learning method that combines the predictions of multiple decision trees and can capture non-linear patterns and interactions in the data [6]. Comparing these algorithms under the same experimental conditions helps evaluate their suitability for a particular dataset. The findings of earlier studies provide background for the present work but should not be confused with the experimental results obtained in this project [1]–[3], [7], [8].

The present project, titled Medical Insurance Cost Predictor, aims to develop a supervised machine-learning approach for estimating medical insurance charges from selected input details. The project uses historical insurance records to train regression models that learn relationships between input features and the target variable. The overall workflow includes dataset inspection, data preprocessing, exploratory data analysis, model development, testing, and performance evaluation. The implementation uses Python-based machine-learning tools, including Scikit-learn, to support data preparation, model training, and evaluation [5]. The dataset source and its characteristics are important for understanding

the scope and reliability of the resulting predictions [4].

Data preprocessing is an essential stage in developing a reliable prediction model. The available records must be examined for missing values, inconsistent entries, data types, and categorical attributes. Numerical features can be used in their existing numerical form, while categorical features such as sex, smoking status, and region need to be converted into a numerical representation before they can be processed by the algorithms. OneHotEncoder is one of the preprocessing techniques available for representing categorical variables [14]. Appropriate preprocessing ensures that the training and testing data are handled consistently. Data transformations should be fitted using the training data and then applied to the test data to avoid data leakage and produce a more reliable evaluation [15].

Exploratory Data Analysis (EDA) is performed to understand the structure, distribution, and relationships within the datasets before applying machine-learning algorithms. Descriptive statistics and graphical visualizations help identify patterns and provide an overview of the available features. A histogram can be used to examine the distribution of medical insurance charges, while a scatter plot can show the relationship between age and charges. A box plot can help compare insurance charges across different categories, such as smoking status. A correlation heatmap can provide an overview of relationships among numerical variables. These visualizations help understand the data and support the selection and interpretation of features used during model development.

This project considers two supervised regression algorithms: Linear Regression and Random Forest Regression. Linear Regression establishes a baseline by estimating insurance charges using a linear relationship between the input features and the target variable. Random Forest Regression uses multiple decision trees and combines their outputs to generate a final prediction. The two algorithms are trained and evaluated to examine their performance on the selected datasets. Their comparison helps identify differences in prediction errors and shows how model performance can vary depending on the characteristics and size of the available data.

The performance of the developed models is evaluated using four standard regression metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MAE measures the average absolute difference between actual and predicted values. MSE calculates the average squared prediction error and gives greater importance to larger errors. RMSE is the square root of MSE and expresses the prediction error in the same unit as the target variable. R^2 indicates how much of the variation in the target variable is explained by the model relative to a baseline prediction. Using these metrics together provides a more complete understanding of model performance than depending on a single evaluation measure.

In addition to model development, the project includes a Streamlit-based dashboard to demonstrate the application-oriented aspects of the system. A graphical interface allows users to interact with the application through input fields and displayed information rather than directly executing machine-learning code. The dashboard is intended to make the project easier to demonstrate and understand. Previous work has also explored application-oriented systems for medical insurance cost prediction [3], [9]. The dashboard and the regression experiments represent related components of the project; however, the prediction models should be described as integrated into the dashboard only if the trained models are actually connected and invoked by the application.

The project also highlights the importance of selecting suitable datasets and interpreting model results carefully. The number of records, quality of input features, distribution of medical charges, and selection of training and testing samples can influence prediction performance. A model producing an output does not automatically mean that it is suitable for practical insurance decisions. Its performance must be examined using appropriate evaluation metrics and, ideally, validated on a larger and more representative dataset. Therefore, the experimental results should be interpreted according to the available data rather than generalized to all individuals or insurance providers.

The project should not be interpreted as a clinical assessment, an official insurance quotation, or a guarantee of premium eligibility. Predictions reflect patterns learned from the training data and may not generalize to other countries, insurers, policy terms, or populations. The commonly used Medical Cost Personal Dataset contains United States regional categories and medical charges in its source currency [4]. Therefore, results obtained from this dataset cannot directly establish accurate medical insurance costs for Indian regions or Indian-rupee values. An India-focused prediction system would require a suitable Indian dataset, clearly defined target units, and model retraining using relevant local records. Any illustrative treatment-cost information should also be distinguished from verified hospital pricing.

The main objectives of this project are to study the factors associated with medical insurance charges, inspect and preprocess the available datasets, perform exploratory data analysis, implement Linear Regression and Random Forest Regression, and compare their predictive performance using standard evaluation metrics. Another objective is to demonstrate the project through a user-friendly Streamlit dashboard and present the application of machine-learning techniques in insurance-cost estimation. The project also aims to identify limitations in the available data and suggest possible improvements for future development.

Overall, the Medical Insurance Cost Predictor provides an educational and experimental study of applying supervised machine-learning techniques to medical insurance cost estimation. It combines data preprocessing, exploratory analysis, regression modelling, performance evaluation, and dashboard development within a single project. By comparing different regression algorithms and examining the limitations of the datasets, the study provides an understanding of how machine learning can be applied to insurance-cost prediction and what additional work is required before such a system can be considered reliable for real-world use.

II . LITERATURE SURVEY

Orji and Ukwandu studied explainable medical-insurance cost prediction using Random Forest, Gradient Boosting Machine, and XGBoost. They evaluated the models with R^2, MAE, RMSE, and MAPE and used SHAP and Individual

Conditional Expectation plots to interpret influential variables. Their work motivates both model comparison and the need to explain the factors associated with predicted premiums [1].

Wang compared Linear Regression and Random Forest on the Medical Cost Personal Dataset. The paper reports that Random Forest outperformed the linear baseline in that particular experiment and identifies smoking status, BMI, and age as important predictors. These results provide a relevant comparison point, but they must not be copied as the performance of the present implementation [2].

Telis, Bhat, and Nausheeda describe a Random Forest-based insurance-cost prediction pipeline with preprocessing, feature scaling, categorical encoding, and a Streamlit interface. Their work is relevant to the proposed interactive dashboard and demonstrates an application-oriented implementation [3].

TABLE I. COMPARISON OF RELATED STUDIES

Study	Dataset / Scope	Methods	Key contribution	Limitation / relevance
Orji & Ukwandu [1]	986 insurance records	RF, GBM, XGBoost; SHAP, ICE	Compares ensemble models and interpretability methods	Different sample size and experiment; results are not transferable directly
Wang [2]	Kaggle Medical Cost Personal Dataset	Linear Regression, Random Forest	Compares baseline and tree ensemble; discusses feature importance	Uses US oriented dataset and should be independently reproduced
Telis et al. [3]	Insurance attributes and policy information	Random Forest, preprocessing, Streamlit	Demonstrates user-facing prediction interface	Dataset and currency assumptions must be checked for local use

III. PROBLEM STATEMENT AND RESEARCH GAP

A. Problem Statement

Medical insurance costs vary among individuals based on several factors, such as age, gender, Body Mass Index (BMI), smoking habits, number of children, and geographical region. Estimating medical insurance charges accurately is challenging because these factors may influence the overall cost differently for each individual.

Traditional methods of estimating insurance costs may not effectively capture the relationships between multiple influencing factors. Therefore, there is a need for a data-driven approach that can analyze medical insurance data and provide cost estimates.

The proposed Medical Insurance Cost Predictor addresses this problem by applying machine learning regression techniques to estimate medical insurance charges. Linear Regression and Random Forest Regression are implemented to analyze the dataset and compare their prediction performance using standard evaluation metrics.

B. Research Gap

Based on the review of existing literature, the following research gaps have been identified:

1. Different machine learning algorithms have been explored for medical insurance cost prediction, but their performance may vary depending on the dataset.
2. There is a need to evaluate and compare multiple regression algorithms using common evaluation metrics.
3. Understanding the relationship between personal characteristics and insurance charges is important for improving cost estimation.
4. There is a need for a simple and interactive web application that makes insurance cost prediction accessible to users.

To address these gaps, the proposed project implements Linear Regression and Random Forest Regression, evaluates their performance using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score, and develops a Streamlit-based web application to demonstrate medical insurance cost prediction.

IV. OBJECTIVES

The primary objective of the proposed Medical Insurance Cost Predictor is to develop a machine learning-based approach for estimating medical insurance charges using available personal and demographic information. The specific objectives of this project are as follows:

1. To collect and analyze a structured medical insurance dataset containing relevant personal, demographic, and lifestyle-related attributes.
2. To identify the important factors associated with medical insurance charges, such as age, sex, Body Mass Index (BMI), number of children, smoking status, and residential region.
3. To perform data preprocessing by examining missing values, duplicate records, data types, and categorical variables to improve data quality.
4. To apply appropriate data encoding techniques, such as One-Hot Encoding, to convert categorical attributes into numerical representations suitable for machine learning algorithms.
5. To perform Exploratory Data Analysis (EDA) using statistical methods and visualizations to understand data distributions, patterns, and relationships among different attributes.
6. To implement Linear Regression as a baseline machine learning algorithm for estimating medical insurance charges.
7. To implement Random Forest Regression to capture potential non-linear relationships and interactions among input features.
8. To evaluate and compare the performance of both regression models using Mean Absolute Error (MAE),

Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

9. To develop an interactive web application using Streamlit that demonstrates the medical insurance cost prediction workflow and presents relevant information through a user-friendly interface.
10. To identify the limitations of the available dataset and model performance and suggest improvements for developing a more reliable medical insurance cost prediction system.

V. SYSTEM ARCHITECTURE

The system architecture of the Medical Insurance Cost Predictor describes the overall workflow, starting from dataset collection and preprocessing to model development, evaluation, and web application development. The system uses machine learning regression algorithms to estimate medical insurance costs based on personal and demographic information.

The proposed architecture consists of the following stages:

1. Dataset Collection: The medical insurance dataset is collected with features such as age, sex, BMI, number of children, smoking status, region, and insurance charges.

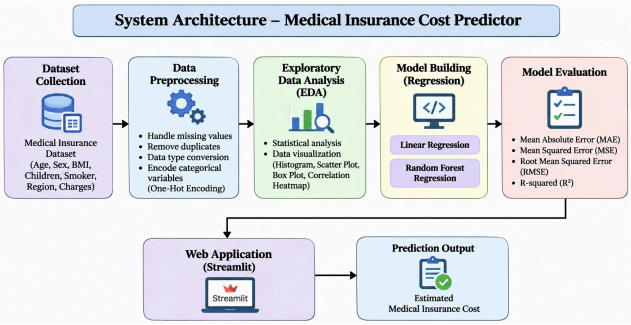
2. Data Preprocessing: The dataset is checked for missing values, duplicate records, and incorrect data types. Categorical variables are converted into numerical values using encoding techniques.

3. Exploratory Data Analysis (EDA): Data analysis and visualizations are performed to understand the distribution of insurance charges and the relationship between different input features.

4. Model Training: Two regression algorithms, Linear Regression and Random Forest Regression, are implemented using the prepared dataset.

5. Model Evaluation: The models are evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2).

6. Web Application: A Streamlit-based web application provides a user interface to demonstrate the project's insurance-related features. The trained regression models and the current demonstration application are separate components unless model integration is completed.



VI. DATASET DESCRIPTION AND PREPROCESSING

The commonly used Medical Cost Personal Dataset distributed through Kaggle contains 1,338 records and seven columns: age, sex, BMI, children, smoker, region, and charges [4], [10], [11]. The six descriptive attributes are candidate predictors, while charges is the regression target. The dataset is widely used for teaching regression and insurance cost analysis; however, its region field represents four United States areas (northeast, northwest, southeast, and southwest), not Indian cardinal regions [4].

A. Feature Description

TABLE II. DATASET ATTRIBUTES

Attribute	Type	Role	Description
age	Integer	Input	Age of primary beneficiary, in years
sex	Categorical	Input	Recorded sex category in source data
bmi	Float	Input	Body mass index (kg/m²)
children	Integer	Input	Number of covered dependents
smoker	Categorical	Input	Smoking status (yes/no)
region	Categorical	Input	Residential region as defined by source dataset
charges	Float	Target	Individual medical cost billed in source dataset currency

B. Preprocessing

Preprocessing begins by checking the column types, missing entries, repeated rows, and summary statistics. Any duplicate or incomplete records are reviewed before a decision is made to remove or retain them. Text-based predictors are converted into
numeric indicator columns through One-Hot Encoding [14]. Numeric columns are kept in their original scale or standardized when the selected model benefits from it. The preprocessing steps are fitted using the training data only, which helps prevent information from the test set from leaking into model development [15]. The planned split assigns 80% of the records to training and 20% to testing, using a fixed random seed.

VII . EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand the dataset and identify patterns, trends, and relationships among the different variables. It helps examine the distribution of medical insurance charges and understand how factors such as age, BMI, smoking status, and other features are related to insurance costs.

Different visualization techniques are used for data analysis. A histogram is used to observe the distribution of insurance charges and identify skewness. A box plot is used to compare insurance charges between smokers and non-smokers and identify possible outliers. A scatter plot can be used to study the relationship between age and insurance charges. A correlation heatmap helps identify the relationships among numerical variables.

EDA plays an important role in understanding the dataset before applying machine learning algorithms. The insights obtained from these visualizations help in selecting relevant

features and preparing the data for model training and evaluation.

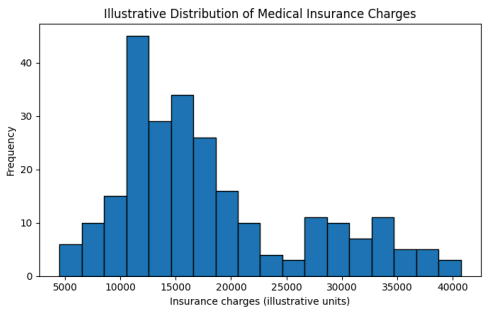


Fig. 1: Illustrative Distribution of Medical Insurance Charges.

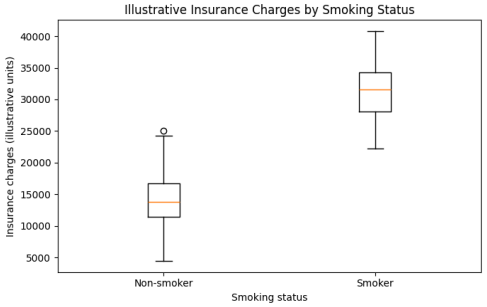


Fig. 2: Comparison of Insurance Charges Between Smokers and Non-Smokers.

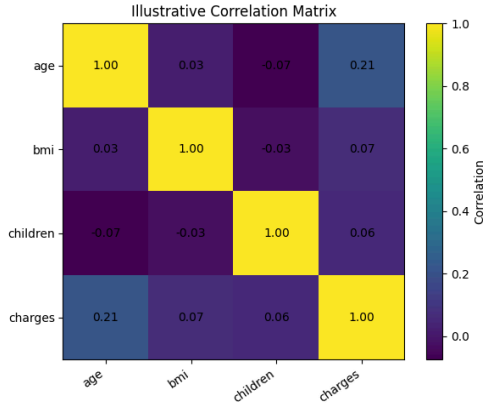


Fig. 3: Correlation Matrix of Dataset Features.

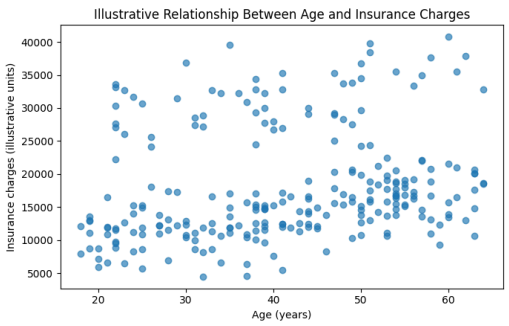


Fig. 4: Relationship Between Age and Medical Insurance Charges.

VIII. PROPOSED METHODOLOGY

The system begins with a structured dataset. After validation and preprocessing, the data is separated into predictor matrix X and target vector y. The training subset is used to fit candidate regression pipelines. The test subset is kept separate until evaluation. Once a model is selected, a dashboard collects the same predictor fields, converts them into a one-row data frame with the expected column names, and sends that row to the saved pipeline. The predicted value is then displayed with a note that it is an estimate.

A. Candidate Algorithms

Two regression approaches are considered in this work. Linear Regression acts as the baseline because it gives a straightforward estimate of how the input variables relate to insurance charges. Earlier studies have also examined linear models alongside tree
based and other machine-learning methods for cost prediction [7], [8], [10]–[12]. The second approach is Random Forest Regression, which combines the predictions of several decision trees and can represent non-linear patterns and interactions between features [6]. The final choice should depend on measured test performance rather than assuming in advance that one model will be superior.

B. Evaluation Metrics

MAE is the mean absolute difference between actual and predicted charges. MSE is the mean squared prediction error. RMSE is the square root of MSE and is expressed in the same unit as the target. R^2 measures the proportion of target variance explained by the model relative to a mean-baseline predictor. Lower MAE, MSE, and RMSE and a higher R^2 usually indicate better test performance.

TABLE III. MODEL EVALUATION METRICS

Metric	Interpretation	Preferred direction
MAE	Average absolute prediction error	Lower
MSE	Average squared prediction error	Lower
RMSE	Square root of MSE; target units	Lower
R^2	Variance explained relative to mean baseline	Higher

IX.WEB APPLICATION AND RESULTS

The Medical Insurance Cost Predictor web application was successfully developed using the Streamlit framework. The application provides a simple and user-friendly interface that allows users to access different insurance-related features. The first page of the application is the login page, where users are asked to enter their full name and 10-digit mobile number to access the dashboard. This is a demonstration login system, and the mobile number is not verified through OTP.

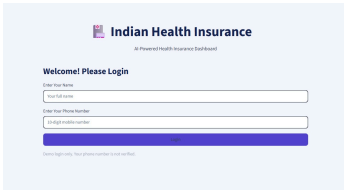
After successful login, the application displays the Home Dashboard, which provides an overview of the user's insurance information. It displays the total coverage, approved claims, and available balance. In the displayed example, the total insurance coverage is ₹5,00,000, the approved claims amount is ₹0, and the available balance is ₹5,00,000. The dashboard also provides navigation options such as AI Disease Predictor, Treatment Cost Estimator, My Insurance Policy, Insurance Balance, Claim History, and Disease Analytics.

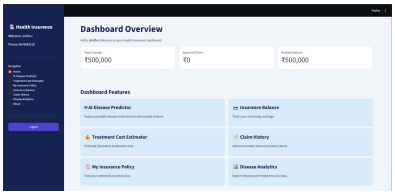
The AI Disease Predictor allows users to enter their health

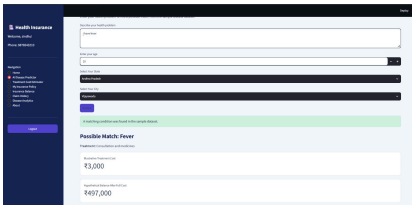
problem description, age, state, and city. In the tested example, the user entered "I have fever", selected the age as 20, and selected Andhra Pradesh and Vijayawada as the location. After clicking the Analyze button, the application identified Fever as a possible match from the sample disease dataset. It displayed consultation and medicines as the suggested treatment. The Treatment Cost Estimator showed an illustrative treatment cost of ₹3,000 and a hypothetical remaining balance of ₹4,97,000 after deducting the estimated cost from the available coverage.

The results demonstrate that the application provides an interactive platform for viewing insurance information, identifying possible disease matches from sample data, and estimating illustrative treatment costs. The application also provides separate sections for insurance policy details, balance information, claim history, and disease analytics. The displayed insurance and treatment values are demonstration data and should not be considered actual insurance records or verified medical costs. The disease predictor uses basic keyword matching and is not a medical diagnosis system.

Overall, the developed web application demonstrates the basic functionality of the Medical Insurance Cost Predictor project through a simple and accessible interface. In future development, the application can be improved by integrating the trained Linear Regression and Random Forest Regression models, using verified medical cost datasets, and implementing secure user authentication.







X . CONCLUSION

This paper presents a structured regression-based approach for medical insurance cost estimation. The proposed pipeline combines data inspection, preprocessing, exploratory analysis, model comparison, evaluation, and an interactive dashboard. Linear Regression provides a baseline, while Random Forest Regression offers a flexible non-linear alternative. The reliability of the final system depends on using a representative dataset, preventing data leakage, and reporting genuine held-out evaluation results. In particular, an India-focused implementation requires Indian insurance data rather than relabeling US regions. Once validated with suitable local data, the system can serve as an educational prototype for data-driven insurance cost estimation. The additional validation, preprocessing, experimental-design, dashboard, and risk tables provide a practical checklist for implementing the system in a reproducible and responsible manner. The present manuscript distinguishes the proposed workflow from completed experimental findings: numerical model performance must be populated only after executing the pipeline on the final dataset. In particular, an India-focused deployment requires a verified Indian dataset and a clearly defined INR target rather than relabelling the US-oriented public dataset.

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